

Visualizing one categorical
variable

Introducing ggplot

Visualizing one quantitative
variable

Describing your distribution:
shape, center, spread?

L03-Visualizing data

Today's objectives

Visualizing one categorical
variable

Introducing ggplot

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Describing your distribution:
shape, center, spread?

Data visualization with ggplot

1. bar chart for categorical variable
2. histogram for continuous variable
 - ▶ describe visualized distributions based on shape, center, spread

Choosing a visualization

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Describing your distribution:
shape, center, spread?

- ▶ Let's say we are interested in describing the race-ethnic distribution for the population of California
- ▶ What type of a variable is this?
- ▶ How might we go about visualizing this information?

**Visualizing one categorical
variable**

Introducing ggplot

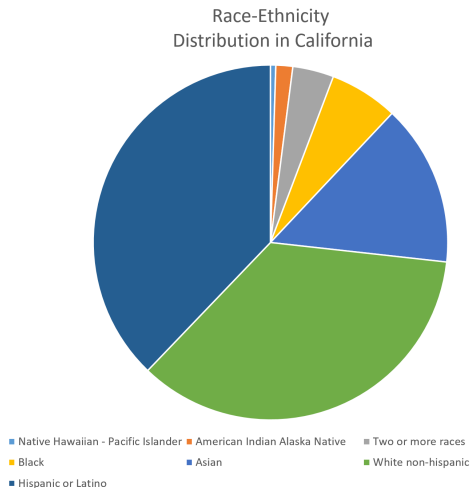
Visualizing one quantitative
variable

Describing your distribution:
shape, center, spread?

Visualizing one categorical variable

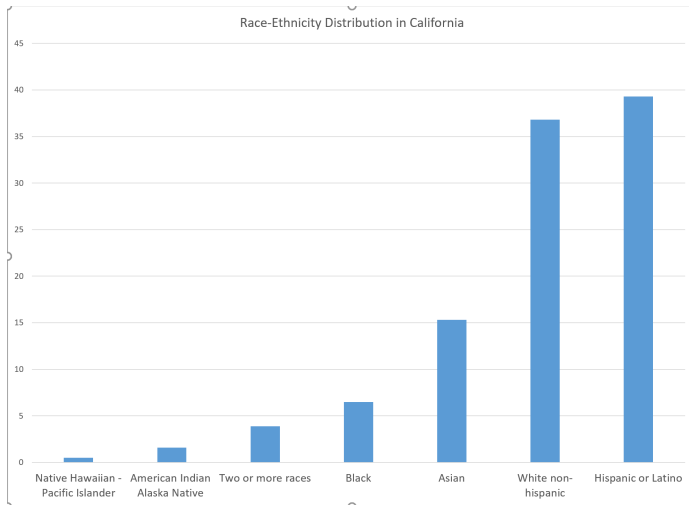
Visualization of categorical data

Generally speaking, it is not a good idea to use pie charts



Can you tell the area of the slices?

Visualization of categorical data



Visualizing one categorical variable

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Describing your distribution: shape, center, spread?

Visualization of categorical data

- ▶ We prefer **bar graphs** (also called **bar charts**) for the display of categorical data.
- ▶ Bar charts display the number or percent of data for each level of the categorical variable being plotted

Example: infectious disease data

- ▶ Task: Make a bar chart of the percent of cases on infectious disease for each category of disease.
- ▶ First, read and view the infectious disease data from Baldi and Moore:

```
id_data <- read_csv("Ch01_ID-data.csv")
```

```
## Rows: 7 Columns: 4
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (2): disease, type
```

```
## dbl (2): number_cases, percent_cases
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this m
```

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Example: infectious disease data

```
id_data
```

```
## # A tibble: 7 x 4
##   disease          type    number_cases percent_cases
##   <chr>          <chr>         <dbl>         <dbl>
## 1 Chlamydia      STI           174557         66.4
## 2 Gonorrhea      STI           44974          17.1
## 3 Pertussis      Pertussis     11219           4.27
## 4 Campylobacteriosis Foodborne     7919           3.01
## 5 Early syphilis STI           7191           2.74
## 6 Salmonellosis  Foodborne     5361           2.04
## 7 Other          Other         11559           4.40
```

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Describing your distribution:
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Example: infectious disease data

- ▶ Note the variables `number_cases` and `percent_cases`
- ▶ What do you want the bar chart to display?
- ▶ What are the x and y variables for a bar chart?

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Describing your distribution:
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Introducing ggplot

First step to building a `ggplot()`: set up the canvas

- ▶ The first line of code below pulls in the `ggplot` package
- ▶ The second line of code below specifies the data set and what goes on the x and y axes

```
library(ggplot2)
ggplot(id_data, aes(x = disease, y = percent_cases))
```

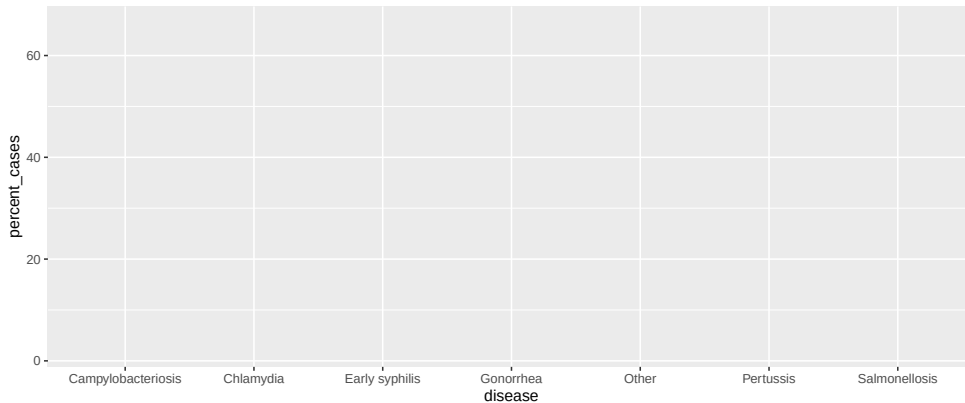
First step to building a `ggplot()`: set up the canvas

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Introducing `ggplot`

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Describing your distribution:
shape, center, spread?



Next choose a function

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Describing your distribution:
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- ▶ We will use a `geom_` function to create our chart

`ggplot()`'s `geom_bar()` makes a bar chart

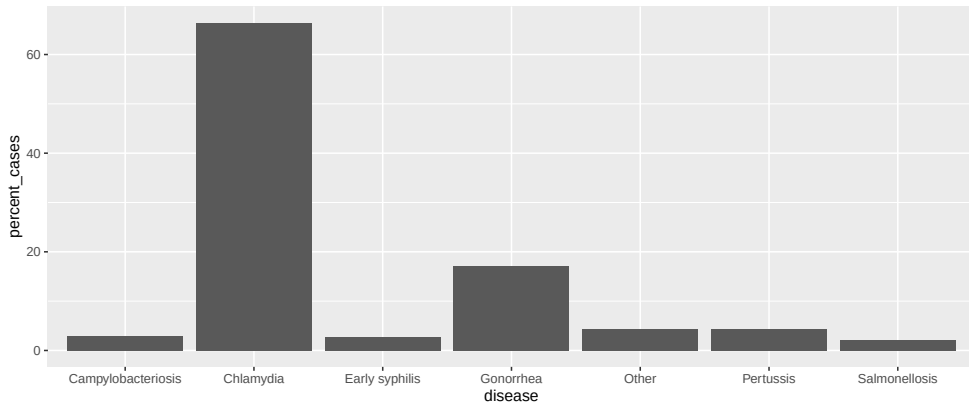
Syntax for bar charts

```
ggplot(id_data, aes(x = disease, y = percent_cases)) +  
geom_bar(stat = "identity")
```

stat = "identity" tells geom_bar that we supplied a y variable that is exactly what we want to plot.

We do not need geom_bar() to calculate the number or percent for us.

ggplot()'s geom_bar() makes a bar chart



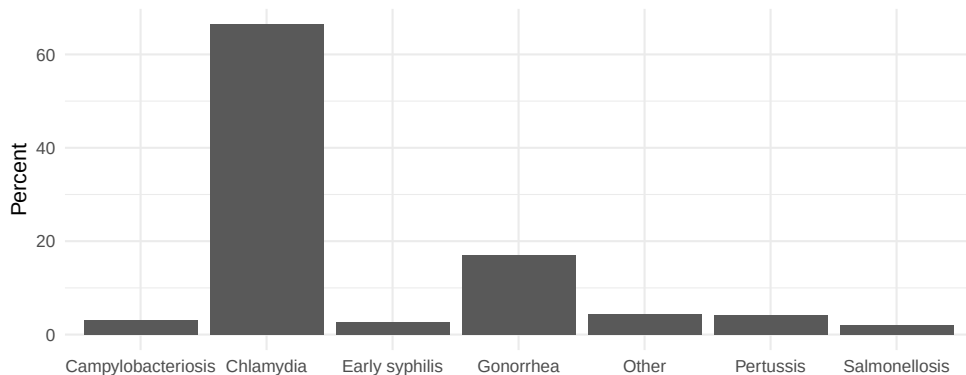
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Describing your distribution:
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some additions to ggplot for style



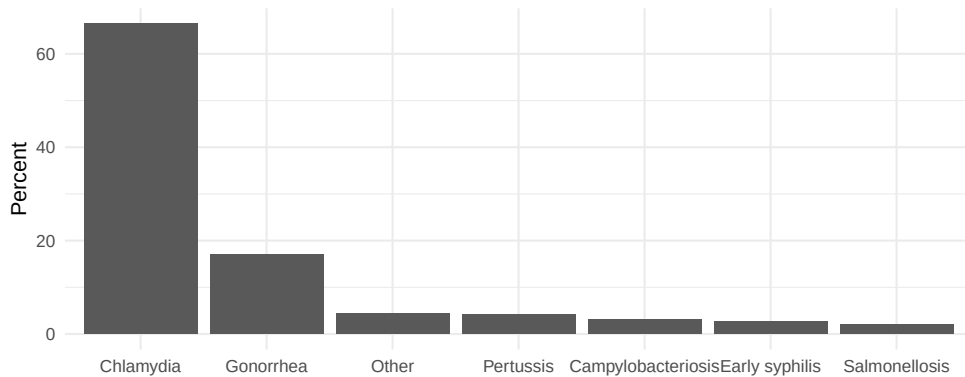
`base_size` controls the font size on these plots

`theme_minimal` affects the “look” of the plot it removes the grey background and adds grey gridlines

fct_reorder reorders disease according to value of
percent_cases

```
library(forcats)  ##notice that we have to pull in the package forcats here!  
id_data <- id_data %>%  
  mutate(disease_ordered = fct_reorder(disease, percent_cases, .desc = T))
```

Re-ordered plot



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Introducing ggplot

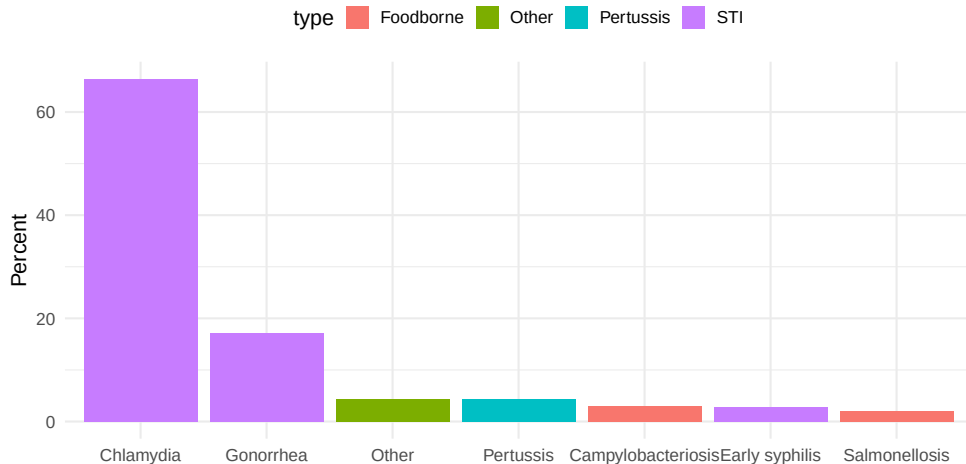
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Describing your distribution:
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Use `aes(fill = type)` to link the bar's fill to the disease type

```
geom_bar(stat = "identity", aes(fill = type)) +  
theme(legend.position = "top")
```

Use `aes(fill = type)` to link the bar's fill to the disease type



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**Visualizing one quantitative
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Describing your distribution:
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Visualizing one quantitative variable

Visualize quantitative variables using histograms

- ▶ Histograms look a lot like bar charts, except that the bars touch because the underlying scale is continuous and the order of the bars matters
- ▶ In order to make a histogram, the underlying data needs to be **binned** into categories and the number or percent of data in each category becomes the height of each bar.
- ▶ the **bins** divide the entire range of data into a series of intervals and counts the number of observations in each interval
- ▶ the intervals must be consecutive and non-overlapping and are almost always chosen to be of equal size

Example: opioid state prescription rates

- ▶ The textbook gives an example using data from 2012.
- ▶ In the data folder, there is updated data from 2018. It came from the paper: “Opioid Prescribing Rates by Congressional Districts, United States, 2016”, by Rolheiser et al. [link](#)

Example: opioid state prescription rates

Problem: To determine the extent to which opioid prescribing rates vary across US congressional districts.

Plan: In an observational cross-sectional framework using secondary data, they constructed 2016 congressional district-level opioid prescribing rate estimates using a population-weighted methodology.

Data: In the data structure we have State as the unit of analysis, and measured perscription rates as the variable of interest

Example: opioid state prescription rates

```
opi_data <- read.csv("Ch01_opioid-data.csv")  
head(opi_data)
```

##	Rank	State	Mean	Median	SD	Min	Max	Num_Districts
## 1	1	AL	121.31	113.09	21.87	105.58	166.69	7
## 2	2	AR	115.22	115.13	8.59	104.80	125.79	4
## 3	3	TN	108.12	108.26	19.16	73.60	133.00	9
## 4	4	MS	105.64	106.25	17.36	83.90	126.14	4
## 5	5	LA	98.38	98.88	10.34	83.22	112.65	6
## 6	6	KY	98.13	85.76	26.72	77.62	147.00	6

- Mean provides the mean prescribing rate per 100 individuals. Thus, a mean of 121.31 implies that in Alabama, there were 121.31 opioid prescriptions per 100 persons, an average across the 7 congressional districts.

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Histogram of opioid prescription rates

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Describing your distribution:
shape, center, spread?

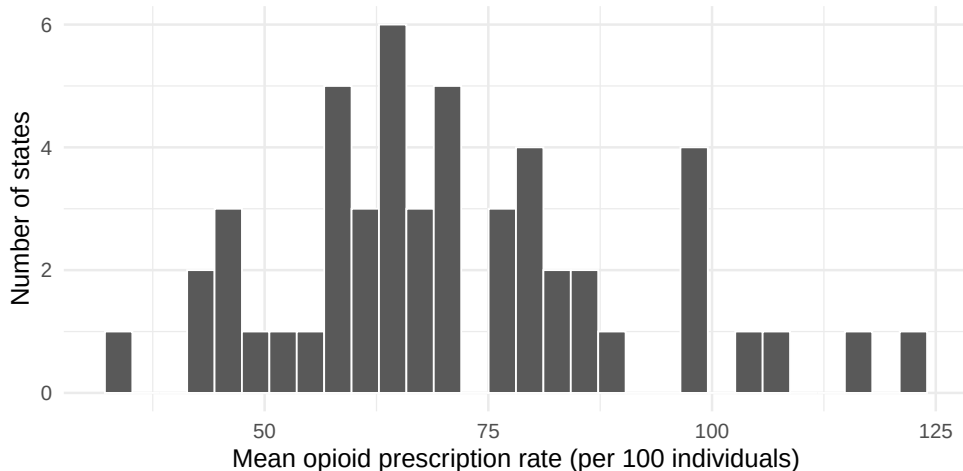
- ▶ Task: Make a histogram of the average prescribing rates across US states
- ▶ What is the x variable? What is the y variable?
- ▶ What geom should be used?

Histogram of opioid prescription rates - default is 30 bins

```
ggplot(data = opi_data, aes(x = Mean)) +  
  geom_histogram(col = "white") +  
  labs(x = "Mean opioid prescription rate (per 100 individuals)",  
        y = "Number of states") +  
  theme_minimal(base_size = 15)
```

Histogram of opioid prescription rates

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'



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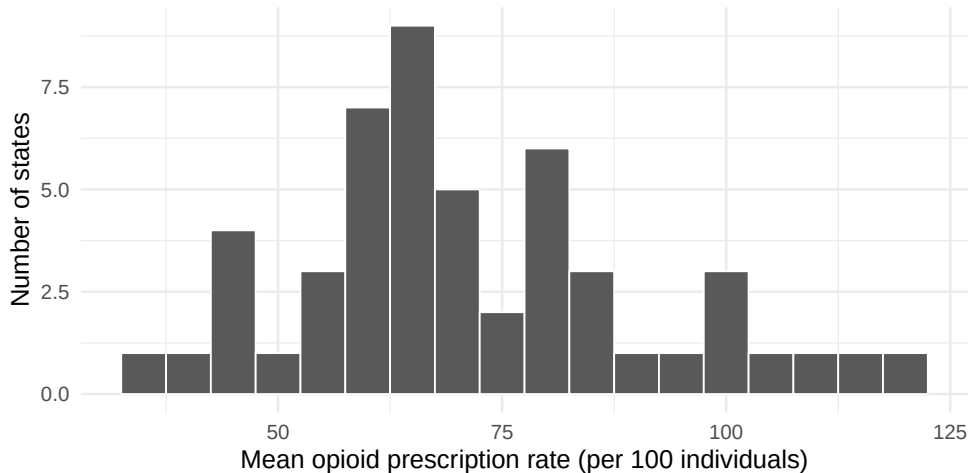
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Describing your distribution: shape, center, spread?

same graph, change the bins `geom_histogram(binwidth = 5)`

```
ggplot(data = opi_data, aes(x = Mean)) +  
  geom_histogram(col = "white", binwidth = 5) +  
  labs(x = "Mean opioid prescription rate (per 100 individuals)",  
        y = "Number of states") +  
  theme_minimal(base_size = 15)
```

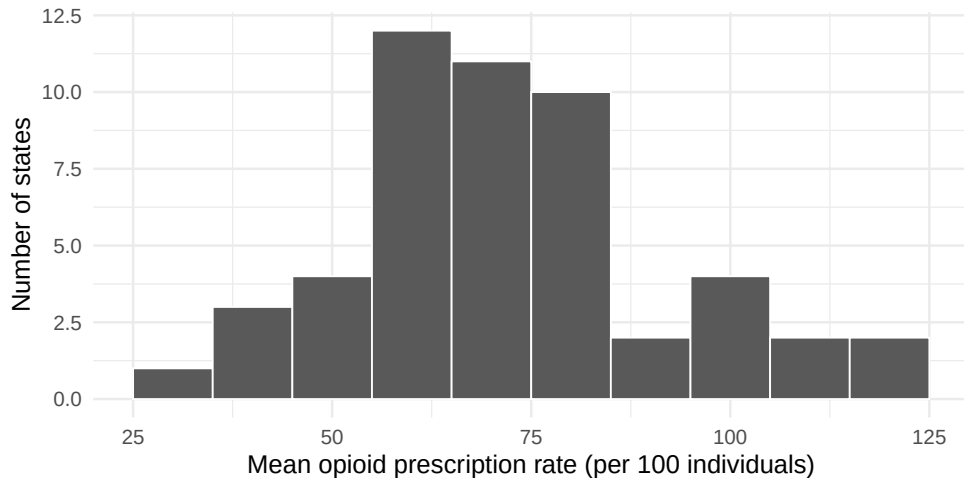
same graph, change the bins `geom_histogram(binwidth = 5)`



change the bins again `geom_histogram(binwidth = 10)`

```
ggplot(data = opi_data, aes(x = Mean)) +  
  geom_histogram(col = "white", binwidth = 10) +  
  labs(x = "Mean opioid prescription rate (per 100 individuals)",  
       y = "Number of states") +  
  theme_minimal(base_size = 15)
```

change the bins again `geom_histogram(binwidth = 10)`



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Describing your distribution:
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Describing your distribution: shape, center, spread?

- ▶ When we examine histograms, we can make comments on a distribution's:
 - ▶ **Shape**: Is the distribution **symmetric** or **skewed** to the left or right?
 - ▶ **Center**: Does the histogram have one peak (unimodal), or two (bimodal) or more?
 - ▶ **Spread**: How spread out are the values? What is the range of the data?
 - ▶ **Outliers**: Do any of the measurements fall outside of the range of most of the data points?

Is this skewed left or skewed right?

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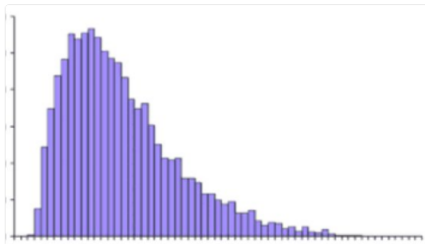
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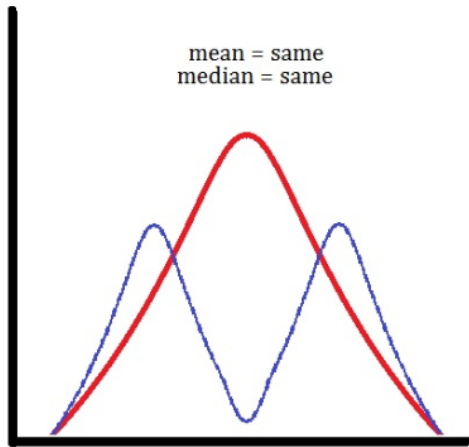
Jesse Singal  @jessesingal · 13h

THIS  IS  NOT  NORMAL 

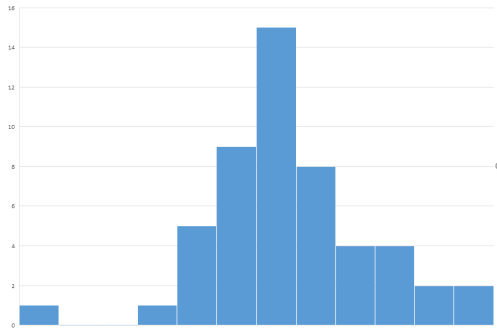


 72  323  1.5K 

Center - one hump or two?



Outlier



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R Recap: Code we used today for visualization

1. 'ggplot' to set up a canvas for graphics
2. `geom_bar(stat = "identity")` to make a bar chart when you specify the y variable
3. `geom_histogram()` to make a histogram for which ggplot needs to calculate the count
4. `fct_reorder(var1, var2)` to reorder a categorical variable (`var1`) by a numeric variable (`var2`)
 - ▶ from the `forcats` package

We only skimmed the surface!

- ▶ Here is some extra material for those of you who love data visualization. This material won't be tested.
 - ▶ RStudio ggplot2 cheatsheet
 - ▶ Kieran Healy's data visualization book

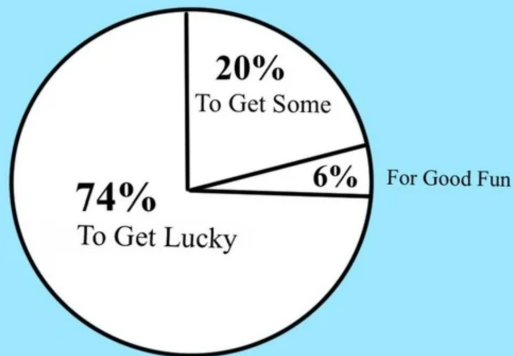
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REASONS WE'RE UP ALL NIGHT



Source: Daft Punk (research assistance by Pharrell Williams)